BILL OF MATERIALS (BOM)

Project: LED Running Light Circuit PCB

Function: Sequential LED chaser using NE555 timer and CD4017 decade counter.

S. No.	Component	Specification / Value	Quantity
1	IC 1	NE555 Timer IC (DIP-8)	1
2	IC 2	CD4017 Decade Counter (DIP-16)	1
3	LED	Red LED (5 mm)	10
4	Resistor	1 kΩ, ¼ Watt	10
5	Resistor R1	10 kΩ, ¼ Watt	1
6	Resistor R2	100 kΩ, ¼ Watt	1
7	Capacitor C1	10 µF, 16V Electrolytic	1
8	Capacitor C2	100 nF Ceramic	1
9	Capacitor C3	100 nF Ceramic	1
10	Power Connector	2-pin DC Input Connector	1
11	PCB	Custom PCB	1

Recommended operating supply: Regulated 5V DC.

Component functions

• NE555 Timer: Generates clock pulses.
• CD4017 Counter: Controls sequential LED operation.
• LEDs: Display the running light effect.
• Resistors: Limit LED current and set the timer frequency.
• Capacitors: Set timing and decouple the IC supply rails.
• Power connector: Provides DC input to the circuit.

Note: Verify component ratings, pin connections and footprints against the selected parts' datasheets before fabrication.